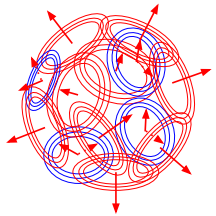
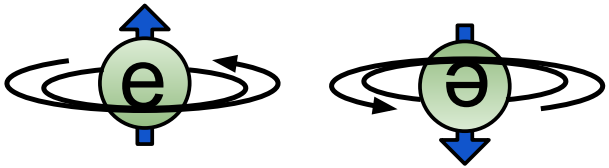
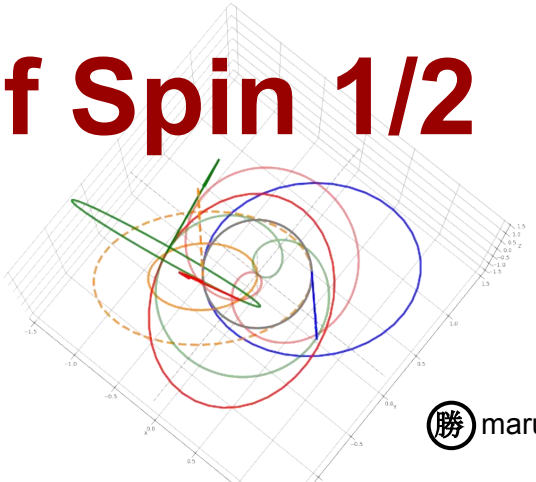
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.3: The Mechanism of Spin 1/2

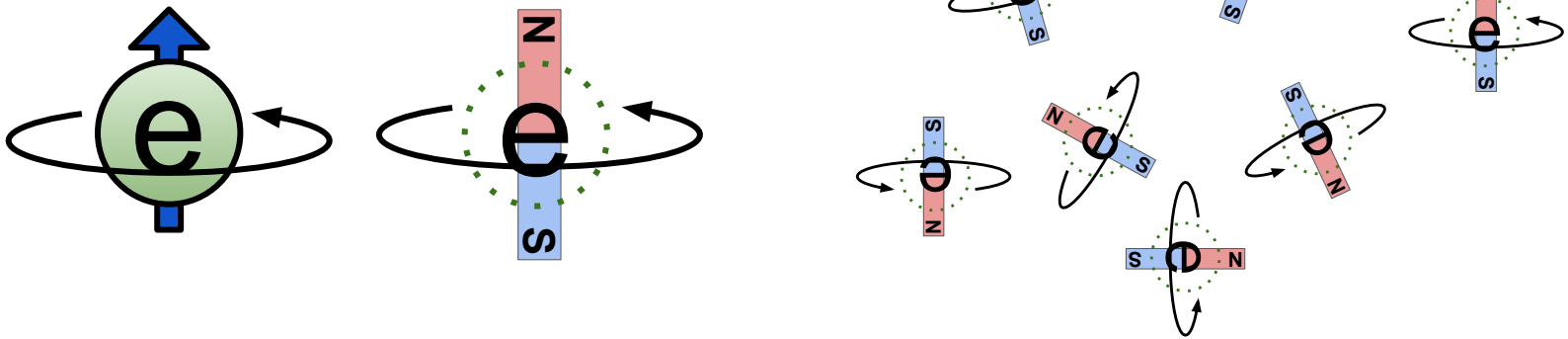




1. Mysterious Properties of Electron Spin

Electrons possess an angular momentum known as spin, and they exhibit magnetic force in the direction of this spin.

An electron can be thought oflf an electron has angular momentum, one would expect every electron to maintain its own various spin directions—like a top maintaining a constant axial orientation.



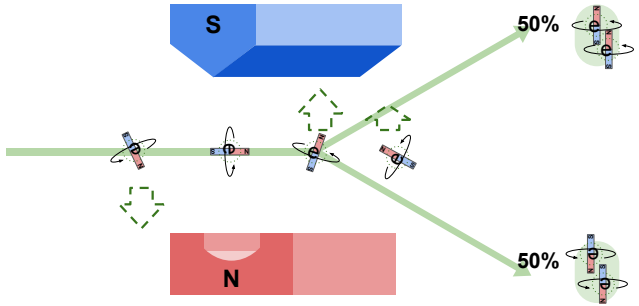
1. Mysterious Properties of Electron Spin

However, when a magnetic field is actually applied to measure the spin direction, it is found that only two directions can ever be measured relative to the direction of the field.

This result remains the same regardless of which direction the magnetic field is applied.

It makes electron spin seem like a physical quantity without a fixed reality that only takes shape upon measurement.

Nevertheless, the reality of magnetic force in magnets is due to this very electron spin.



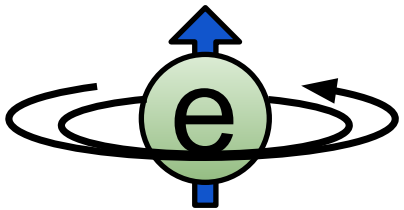
1. Mysterious Properties of Electron Spin

Furthermore, while ordinary objects return to their original orientation after at least one full rotation (360°), an electron only returns to its original state after two full rotations (720°).

This is called 1/2-fold rotational symmetry.

The half-integer "1/2" in "spin 1/2" indicates that the measured spin direction (magnetic direction) has only two possibilities and that it possesses 1/2-fold rotational symmetry.

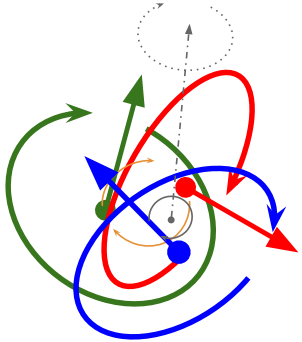
720 degree

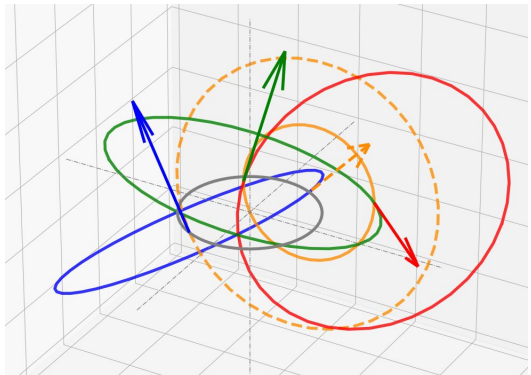


1. Mysterious Properties of Electron Spin

Current physics explains these mysterious characteristics through the SU(2) symmetry of spacetime, but it lacks an answer as to why the electron possesses that specific symmetry.

In R-space Theory, these mysterious properties are explained by the precession and orbital revolution resulting from the combination of the three rotation vectors (angular momentum) that the electron possesses.

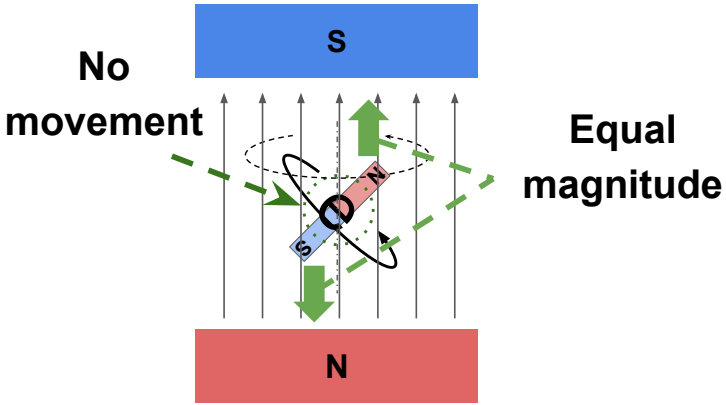




2. The Stern-Gerlach Experiment

Even though electron spin is only measured in two directions relative to the magnetic field, it cannot be measured using a standard, uniform magnetic field.

In a field without a gradient, equal magnetic forces act on the N and S poles of the electron's "magnet," causing it to precess like a top but preventing it from moving in the direction of the applied field.

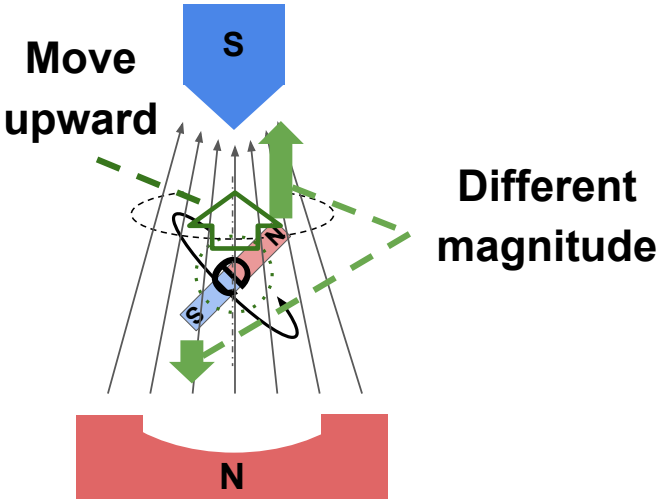


2. The Stern-Gerlach Experiment

In the Stern-Gerlach experiment, a special device was created where the shape of the magnets produces a magnetic field gradient depending on the position.

The distance between the N and S poles relative to the field direction varies based on the orientation of the electron's magnet (the angle of precession).

If there is a gradient, a difference in the magnitude of the magnetic force applied to each pole occurs, causing the electron to move in the direction of the applied magnetic field.

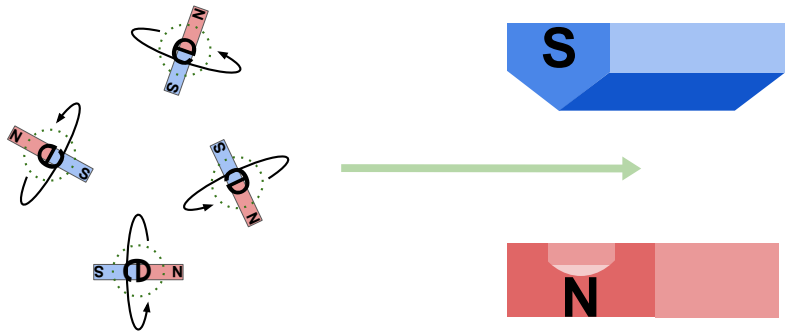


2. The Stern-Gerlach Experiment

First, a magnetic field is applied in the vertical direction, and electrons are passed through it (silver atoms were used in the actual experiment).

Before applying the magnetic force, the orientations of the electrons' magnets are random.

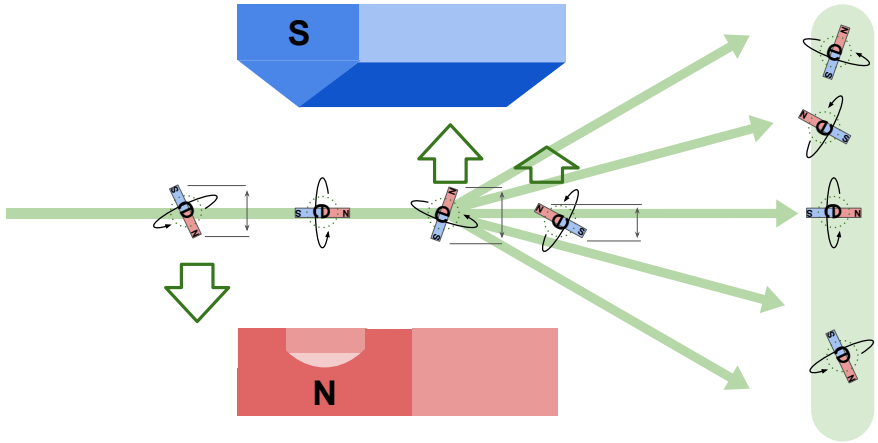
Once the force is applied, the electrons precess like tops, but the angle of that precession should—classically—be random based on the electron's original orientation.



2. The Stern-Gerlach Experiment

If that were the case, the magnitude of the magnetic force would vary depending on the angle of precession and the orientation of the poles.

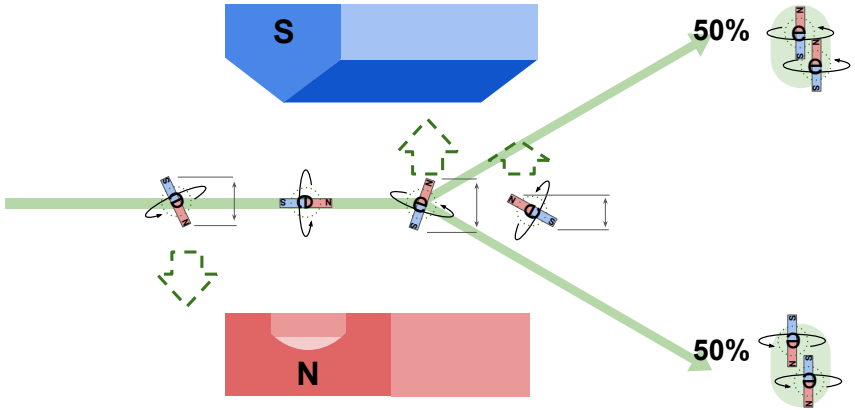
Consequently, electrons should have spread out uniformly between the top and bottom of the observation screen in a fan-like distribution.

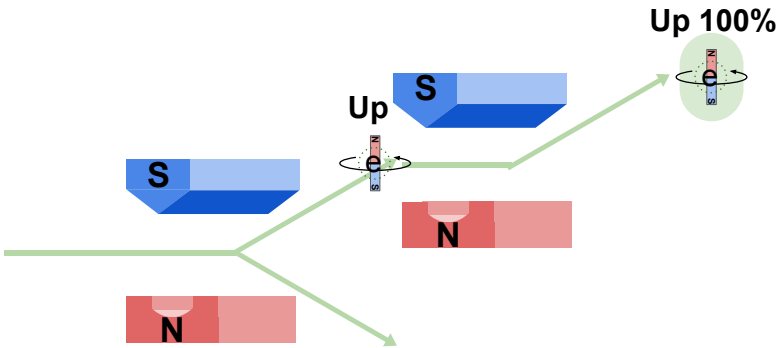


2. The Stern-Gerlach Experiment

However, the experiment showed that electrons split with a 50% probability of being "up" and a 50% probability of being "down," with nothing observed in between.

If electrons measured as "up" are measured again with a vertical magnetic field, they are observed as "up" with 100% probability, suggesting the spin is maintained.

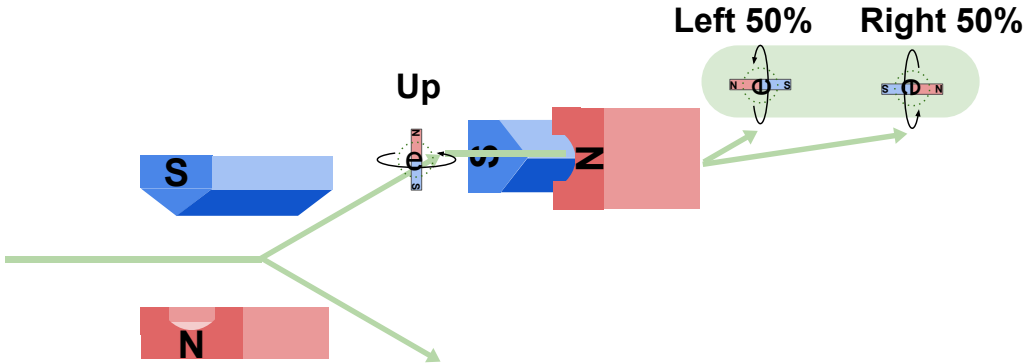




2. The Stern-Gerlach Experiment

Next, if the "up" spins are immediately measured using a horizontal magnetic field, they do not travel straight through unaffected. Instead, the horizontal spin (left/right) is observed with a 50% probability of being "left" and 50% for "right".

Thus, despite being a form of angular momentum, electron spin direction changes depending on how it is measured. R-space Theory explains the mechanism behind this mysterious "electron top".

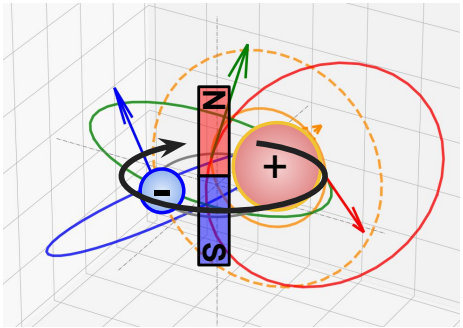


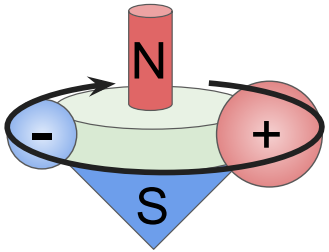
3. Mechanism of Changes in Electron Spin

In R-space, an electron is a vortex of three rotation vectors, where a positive 2/3 charge and a negative1/3 charge revolve around each other.

Because the charges move in a circle, magnetic force is generated like an electromagnet. This is the nature of electron spin in R-space—a "charge top".

Comparing a normal top to this electron top: when a normal top tilts, its center of gravity shifts away from the axis, and gravity acts as a force that tilts the axis further.

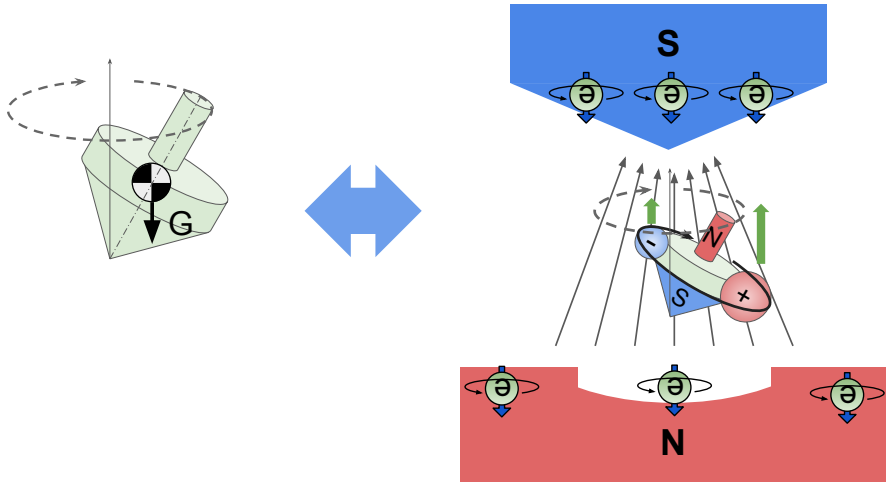




3. Mechanism of Changes in Electron Spin

In the electron top, charges are rotating.

Since magnetic force is a function of charge, the force acts on the electron top's charges, serving as the force that tilts the axis.

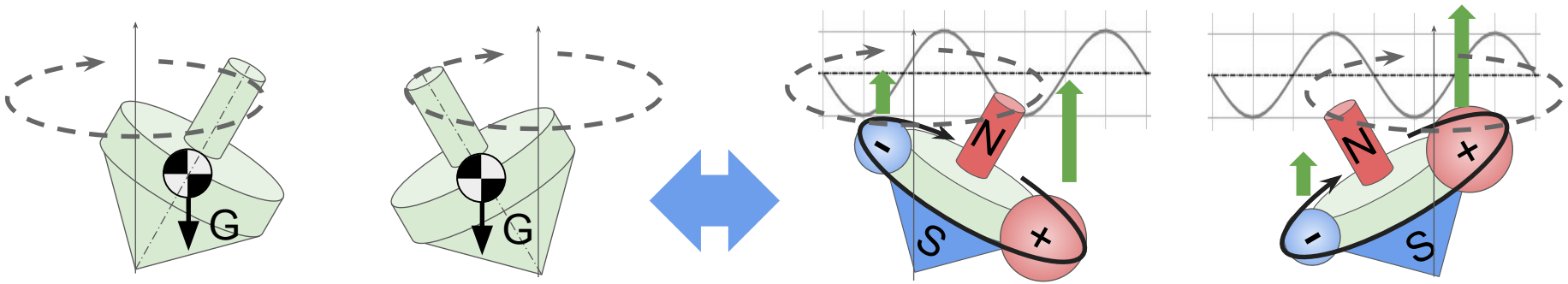


3. Mechanism of Changes in Electron Spin

In a normal top, gravity always acts in the direction the axis is tilted.

However, in the electron top, because the charges are rotating, the force is not always applied in the direction the axis is tilted.

Therefore, the magnitude of the force applied to the axis changes periodically.

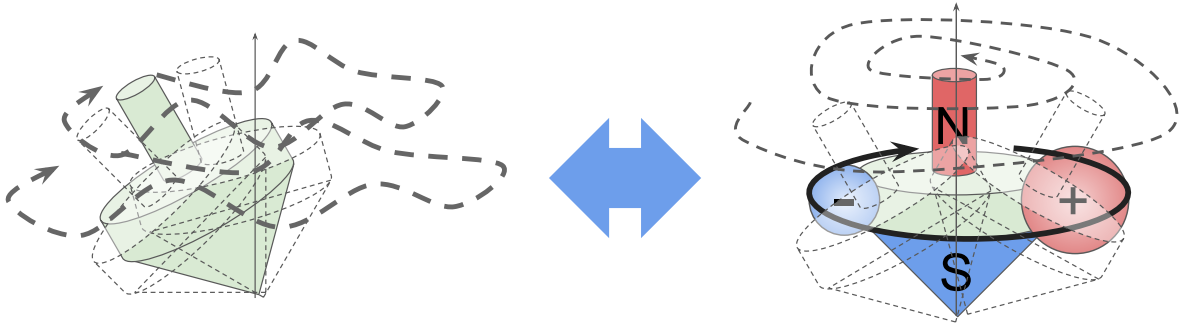


3. Mechanism of Changes in Electron Spin

If you apply a periodically changing force to the axis of a normal top, it begins to nutate—a motion where the angle of precession changes periodically.

If the period of the applied force matches the period of nutation, the precession grows larger, the top becomes unstable, and it falls over.

In contrast, in the electron top, the nutation actually decreases, and the angle of precession becomes smaller. This is because the electron top possesses 1/2-fold rotational symmetry, requiring 720° for one full cycle.

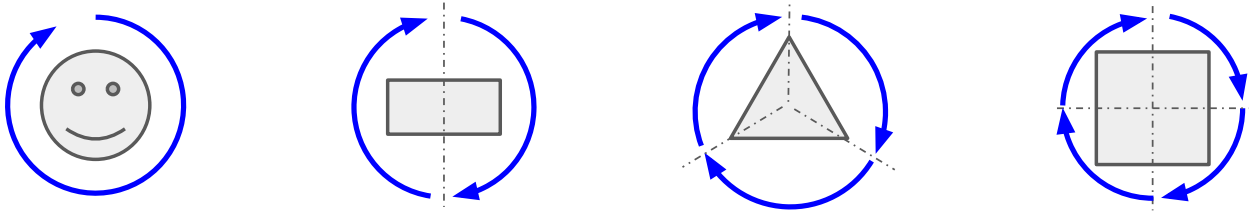


4. 1/2-Fold Rotational Symmetry

Rotational symmetry refers to how many times a shape appears identical during one full 360° rotation.

For an equilateral triangle, rotating it 120° makes it look the same as before; since this happens three times in 360°, it has 3-fold symmetry. A square has 4-fold symmetry.

A complex shape that only looks the same after 360° has 1-fold symmetry.



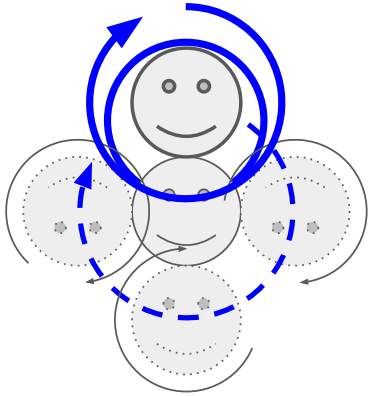
1-fold(360 deg.) 2-fold(180 deg.) 3-fold(120 deg.) 4-fold(90 deg.)

4. 1/2-Fold Rotational Symmetry

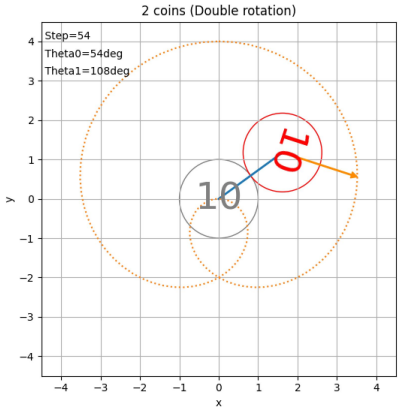
Since 360° / (angle n° to look the same) = n-fold symmetry, 1/2-fold symmetry means it must be rotated 720° to return to its original appearance.

While 1/2-fold symmetry does not exist on a 2D plane, Dirac's belt trick is an example.

Additionally, if two coins are placed in contact and one is rolled around the other, the outer coin rotates 720° to return to its original position, representing 1/2-fold rotational symmetry.



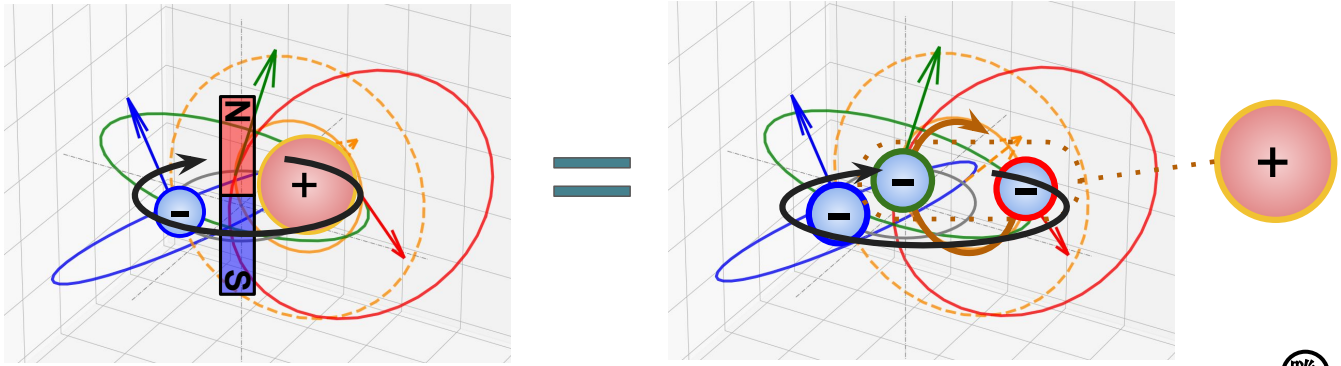
1/2-fold(720 deg.)



5. 1/2-Fold Rotational Symmetry of the Electron Top

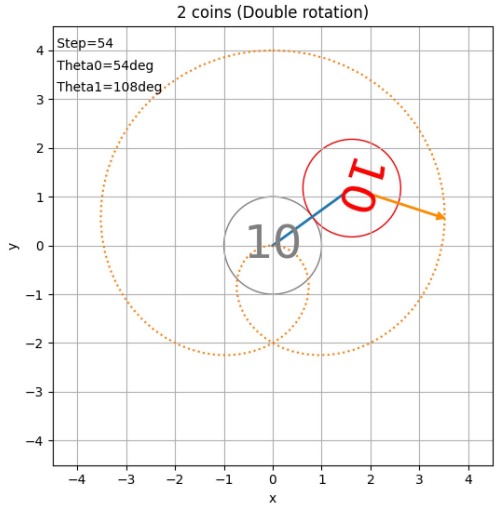
As explained in Chapter 2, of the electron's three rotation vectors, x and y form a pair acting like a positive 2/3 charge, while the remaining z vector acts like a negative 1/3 charge.

The rotation vectors x and y revolve around each other like a binary star system. Individually, however, the rotation vectors x and y are each a negative 1/3 charge according to the charge calculation formula.

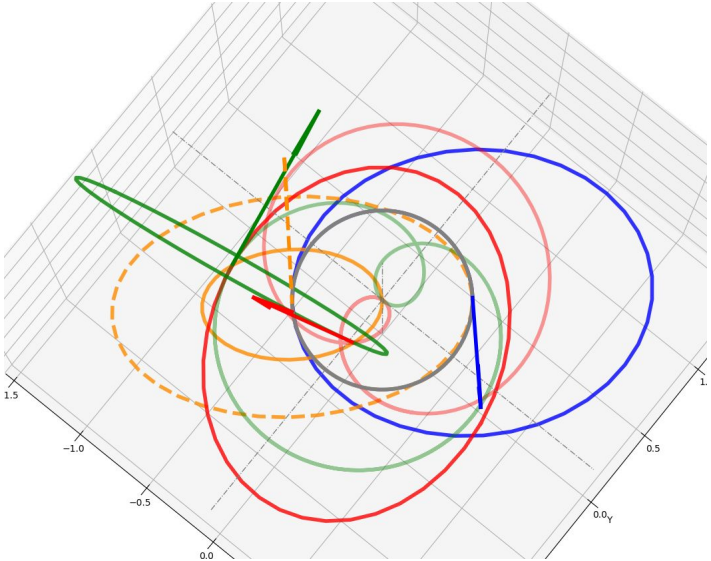


5. 1/2-Fold Rotational Symmetry of the Electron Top

The trajectories of rotation vectors x and y result in a shape identical to the trajectory of the outer coin when two coins are rolled together.



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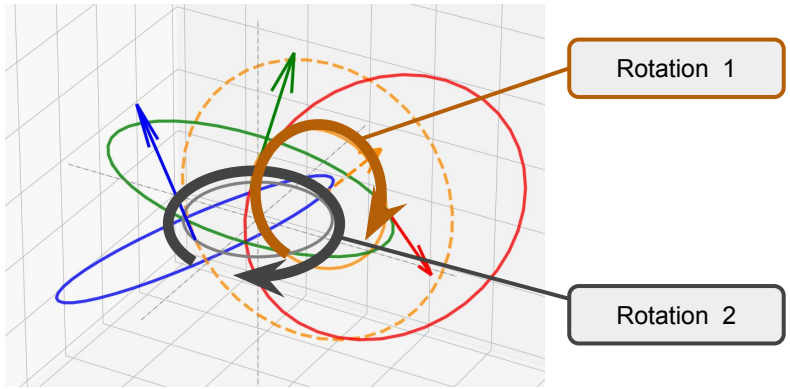


5. 1/2-Fold Rotational Symmetry of the Electron Top

Rolling two coins in contact is a form of "double rotation".

In the electron top, vectors x and y pair up for one orbital revolution (Rotation 1), and this pair then undergoes a second orbital revolution with vector z (Rotation 2), creating a double rotation.

Consequently, the charge of the electron top rotates with 1/2-fold symmetry.



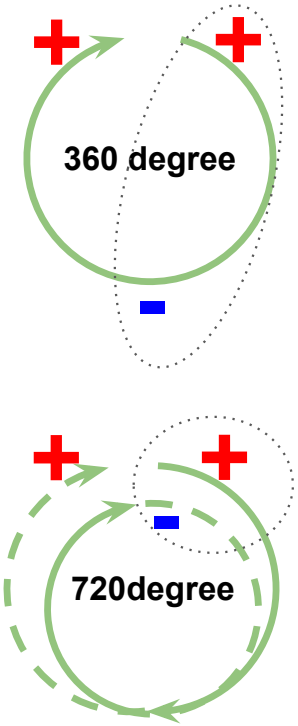
5. 1/2-Fold Rotational Symmetry of the Electron Top

In normal rotation (1-fold symmetry), a 360° turn returns the object to its original state.

If represented by phase signs (+ and -), 0° is (+), 180° is (-), and 360° returns to (+).

In 1/2-fold symmetry, 0° is (+), but 360° is (-), and it only returns to (+) at 720°.

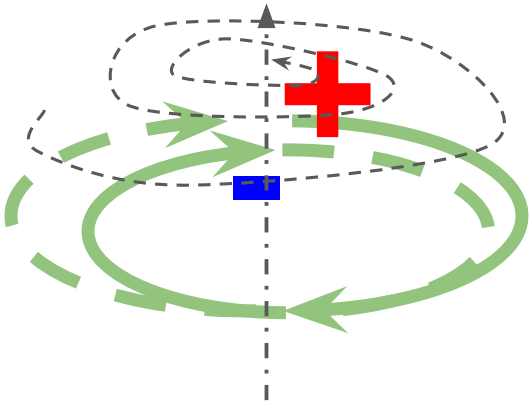
In 1/2-fold symmetry, the sign reverses at 360°.

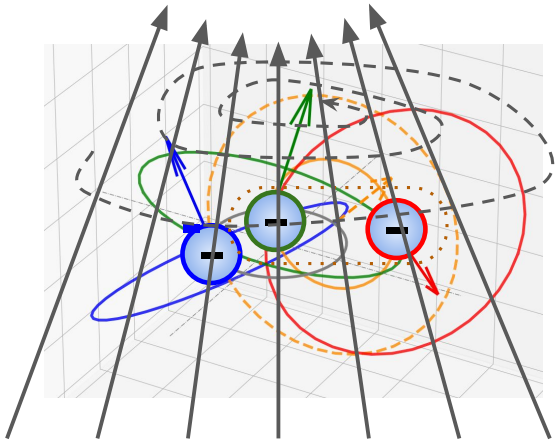


5. 1/2-Fold Rotational Symmetry of the Electron Top

Because of this, the electron top receives magnetic force in a phase opposite to the phase of the spin axis precession.

This acts as a force that cancels out nutation, causing the precession to decrease.

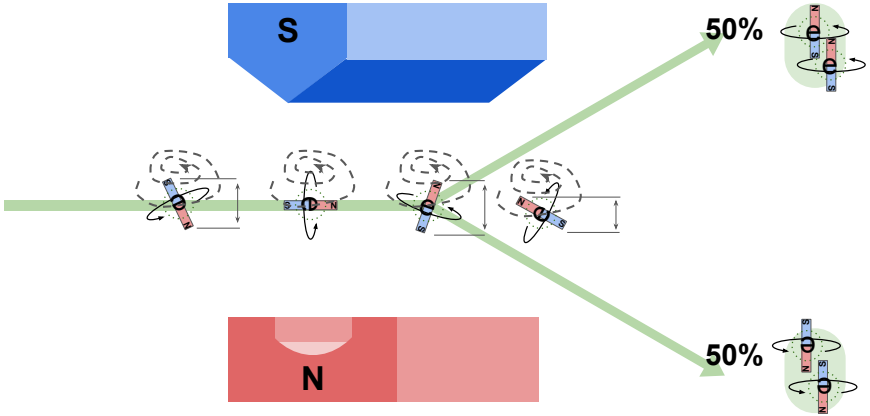




6. Conclusion

The electron top possesses angular momentum, but due to 1/2-fold symmetry, the axis of the top (the axis of angular momentum) aligns with the direction of the applied magnetic field.

Because the electron top rotates at incredible speeds, the axis orientation changes in an extremely short time, resulting in only two directions being observed along the measurement axis.

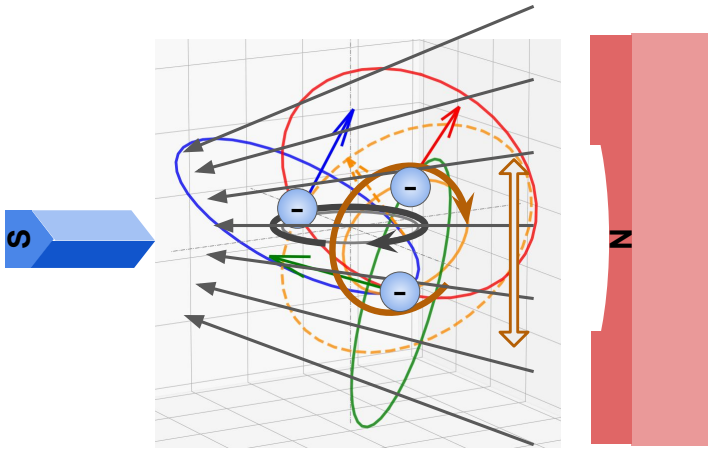


6. Conclusion

Furthermore, the charge of the electron top fluctuates periodically in a direction perpendicular to the plane of rotation.

Therefore, if it is re-measured in an orthogonal direction after being measured once, the periodic force acting on the charge will cause the axis to align either left or right.

Whether it becomes "left" or "right" depends on the timing (phase) of the rotation vectors, making it a 50/50 probabilistic outcome.

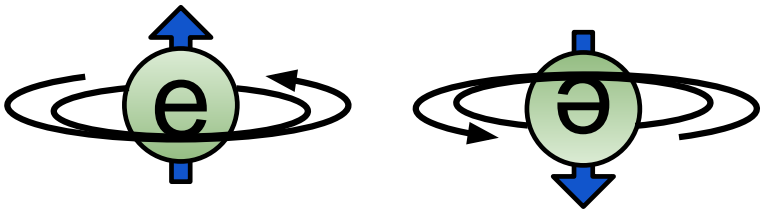


6. Conclusion

This is the reason why electron spin is always observed in only two directions in the Stern-Gerlach experiment, and it is the mechanism behind spin 1/2.

The electron is a top made of vortices of angular momentum within R-space.

Because those vortices undergo double rotation, they react to external forces (magnetic force) differently than normal tops, which is why they appeared so mysterious.



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This is all for my presentation.

Thanks for watching!

